# CE 310 — Week 10 Assignment

Week: 10 | Topic: Multiple Linear Regression — Diagnostics and Application Points: 60 pts total (3 problems)
Due: Wednesday of Week 11 by 9:00 AM — submit to D2L

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## Overview

The lab built multiple regression models adding BusinessHours and Season as predictors alongside Outdoor_Temp_F. This assignment asks you to extend that work: diagnose which predictors actually matter, apply the interaction model to track-specific design problems, and reflect critically on when model complexity is (and is not) worth the trade-off.

You will need: your completed lab notebook (Week10_Lab.ipynb), the DOE dataset (Week5_DOE_MedOffice_ZoneTemp_20
and a Python environment (Google Colab or local).

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## How to Submit

> WARNING: Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

| Requirement | Detail |
|---|---|
| File format | Download as .ipynb — File → Download → Download .ipynb (not PDF, not .py) |
| Major declaration | Set MAJOR = "CE" or MAJOR = "ArcE" in the Setup cell — this determines which Problem 2 track is graded |
| Print lines | Do not edit or delete any print(f"ANSWER_... = ...") line |
| Written responses | Put each in its own markdown cell immediately after the code cell it responds to |
| File name | lastname_firstname_week10.ipynb e.g. smith_jane_week10.ipynb |
| Submission | Upload the .ipynb file to D2L Dropbox before the deadline |

Setup cell (run first):

```
import pandas as pd
import statsmodels.formula.api as smf

MAJOR = "CE"   # or "ArcE" — declares your Problem 2 track

df = pd.read_csv("Week5_DOE_MedOffice_ZoneTemp_2023.csv")
df['BH'] = (df['BusinessHours'] == 'Yes').astype(int)
print(f"MAJOR = {MAJOR}")
```

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## Problem 1 — Model Diagnostics and Extension (30 pts)

Do not repeat the lab computations. Use the model coefficients, $R^2$ values, and residual plots from your completed lab notebook as starting points.

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### P1a (8 pts) — Why Season Adds Almost Nothing

In the lab (Model 3), adding C(Season) alongside Outdoor_Temp_F increased $R^2$ from 0.601 (simple regression) to only 0.611 — a gain of just 1 percentage point. Yet adding BH alone (Model 2) increased $R^2$ from 0.601 to 0.697 — a gain of nearly 10 percentage points.

```python
season_means = df.groupby('Season')['Outdoor_Temp_F'].mean()
ANSWER_P1a_summer_mean = round(season_means['Summer'], 2)
ANSWER_P1a_winter_mean = round(season_means['Winter'], 2)
print(f"ANSWER_P1a_summer_mean = {ANSWER_P1a_summer_mean}")
print(f"ANSWER_P1a_winter_mean = {ANSWER_P1a_winter_mean}")
```

In approximately 75 words, in a markdown cell, explain why Season adds so little predictive power once Outdoor_Temp_F is already in the model. Your explanation must: 1. Use the word collinearity (or collinear). 2. Reference the two numbers you just printed.

Graded on accuracy of the collinearity explanation and correct use of the quantitative example.

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### P1b (8 pts) — DayType as an Alternative Occupancy Proxy

Instead of BusinessHours, fit a model using DayType (Weekday vs. Weekend) as the second predictor:

```python
model_dt = smf.ols('Cooling_kWh ~ Outdoor_Temp_F + C(DayType)', data=df).fit()
ANSWER_P1b_r2 = round(model_dt.rsquared, 4)
ANSWER_P1b_weekend_coef = round(model_dt.params["C(DayType)[T.Weekend]"], 4)
ANSWER_P1b_weekend_pvalue = model_dt.pvalues["C(DayType)[T.Weekend]"]
print(f"ANSWER_P1b_r2 = {ANSWER_P1b_r2}")
print(f"ANSWER_P1b_weekend_coef = {ANSWER_P1b_weekend_coef}")
print(f"ANSWER_P1b_weekend_pvalue = {ANSWER_P1b_weekend_pvalue}")
print(model_dt.summary())
```

In a markdown cell: Is DayType a statistically significant predictor after controlling for outdoor temperature (justify using the p-value)? Compare this model's $R^2$ to Model 2 ($R^2 = 0.697$, which used BH). Which predictor — BusinessHours or DayType — explains more variance in cooling energy? Why might one outperform the other? (Hint: think about what DayType captures vs. what BusinessHours captures at the hourly level.)

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### P1c (7 pts) — Adding Both BH and DayType

Fit a model with both BH and DayType:

```python
model_bh_dt = smf.ols('Cooling_kWh ~ Outdoor_Temp_F + BH + C(DayType)', data=df).fit()
ANSWER_P1c_r2 = round(model_bh_dt.rsquared, 4)
ANSWER_P1c_adj_r2 = round(model_bh_dt.rsquared_adj, 4)
print(f"ANSWER_P1c_r2 = {ANSWER_P1c_r2}")
print(f"ANSWER_P1c_adj_r2 = {ANSWER_P1c_adj_r2}")
print(model_bh_dt.summary())
```

In a markdown cell: Compare to Model 2 ($R^2$ = 0.697, Adj. $R^2$ = 0.697). Does adding DayType on top of BH improve $R^2$? Does Adj. $R^2$ increase or decrease? What do you conclude about the incremental information in DayType once BH is already included? (Hint: consider what BH and DayType each capture — are they measuring similar or different aspects of occupancy?)

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P1d (7 pts) — Interaction Model Predictions at the 0.4% Design Temperature

Using Model 5 (the interaction model Cooling_kWh ~ Outdoor_Temp_F * BH) from the lab, predict the cooling demand at the 0.4% design temperature = 103.8°F for two scenarios:

```
model5 = smf.ols('Cooling_kWh ~ Outdoor_Temp_F * BH', data=df).fit()
b0, b1, b2, b3 = (model5.params['Intercept'], model5.params['Outdoor_Temp_F'],
                  model5.params['BH'], model5.params['Outdoor_Temp_F:BH'])

design_temp = 103.8

# (a) Occupied scenario (BH = 1)
ANSWER_P1d_occ_pred = round(b0 + b1*design_temp + b2*1 + b3*design_temp*1, 2)
print(f"ANSWER_P1d_occ_pred = {ANSWER_P1d_occ_pred}")

# (b) Unoccupied scenario (BH = 0)
ANSWER_P1d_unocc_pred = round(b0 + b1*design_temp + b2*0 + b3*design_temp*0, 2)
print(f"ANSWER_P1d_unocc_pred = {ANSWER_P1d_unocc_pred}")

# Compare to Model 2 prediction at the same temperature
model2 = smf.ols('Cooling_kWh ~ Outdoor_Temp_F + BH', data=df).fit()
ANSWER_P1d_model2_pred = round(
    model2.params['Intercept'] + model2.params['Outdoor_Temp_F']*design_temp + model2.params['BH'], 2
)
ANSWER_P1d_diff = round(ANSWER_P1d_occ_pred - ANSWER_P1d_model2_pred, 2)
print(f"ANSWER_P1d_model2_pred = {ANSWER_P1d_model2_pred}")
print(f"ANSWER_P1d_diff = {ANSWER_P1d_diff}")
```

State the four Model 5 coefficients by name (b0–b3 above) before you substitute — show the full substitution step-by-step in a markdown cell, not just the code.

In the same markdown cell: at the 0.4% design condition, which model would lead to a larger-capacity chiller specification? Explain in one sentence why the interaction model's occupied estimate is higher than the additive model's estimate at high temperatures.

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## Problem 2 — Track Application (18 pts)

Complete only the track matching your declared MAJOR. CE → Track A. ArcE → Track B.

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Track A — CE: Interaction Model for a 24/7 Occupied Facility (18 pts)

A mechanical engineer is sizing a chiller for a data center that operates 24 hours a day, 7 days a week — the building is always "occupied" in our model (BH = 1 at all hours).

(A2-a, 6 pts) Simplify Model 5 algebraically for occupied operation only (BH = 1) to the form $\hat{y} = m \cdot T + b$:

```python
ANSWER_A2a_occ_slope = round(b1 + b3, 4)
ANSWER_A2a_occ_intercept = round(b0 + b2, 4)
print(f"ANSWER_A2a_occ_slope = {ANSWER_A2a_occ_slope}")
print(f"ANSWER_A2a_occ_intercept = {ANSWER_A2a_occ_intercept}")
```

(A2-b, 6 pts) The ASHRAE 90.4 design condition is the 0.4% outdoor dry-bulb temperature = 103.8°F. Using the occupied equation from A2-a, predict cooling demand at this design condition, and compare to a model using the Week 9 all-hours slope (1.1585 kWh/°F, intercept = −60.68 kWh):

```python
ANSWER_A2b_pred_design = round(ANSWER_A2a_occ_slope*103.8 + ANSWER_A2a_occ_intercept, 2)
ANSWER_A2b_wk9_pred = round(1.1585*103.8 + (-60.68), 2)
ANSWER_A2b_diff = round(ANSWER_A2b_pred_design - ANSWER_A2b_wk9_pred, 2)
print(f"ANSWER_A2b_pred_design = {ANSWER_A2b_pred_design}")
print(f"ANSWER_A2b_wk9_pred = {ANSWER_A2b_wk9_pred}")
print(f"ANSWER_A2b_diff = {ANSWER_A2b_diff}")
```

In a markdown cell, in one sentence: explain physically why the interaction model's occupied slope (≈1.78 kWh/°F) is steeper than the all-hours slope (≈1.16 kWh/°F).

(A2-c, 6 pts) The building's chiller has a rated capacity of 200 tons (703.4 kWh/hr). Using the occupied prediction equation from A2-a, solve for the outdoor temperature at which the chiller would reach full capacity:

```python
chiller_capacity = 200 * 3.517
ANSWER_A2c_Tsat = round((chiller_capacity - ANSWER_A2a_occ_intercept) / ANSWER_A2a_occ_slope, 2)
print(f"ANSWER_A2c_Tsat = {ANSWER_A2c_Tsat}")
```

In a markdown cell (2–3 sentences): is this temperature physically plausible for Tucson (max observed ≈106°F)? What does this imply about whether the 200-ton chiller could ever be overwhelmed by outdoor heat alone, assuming all occupancy loads remain at the modeled level?

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Track B — ArcE: ASHRAE 90.1 Occupied/Unoccupied Energy Model (18 pts)

ASHRAE Standard 90.1-2022 requires that energy compliance simulations model occupied and unoccupied operating modes separately. The interaction model gives you separate equations for each mode.

(B2-a, 6 pts) Simplify Model 5 into two separate equations — one for occupied hours (BH = 1), one for unoccupied hours (BH = 0):

```python
ANSWER_B2a_occ_slope = round(b1 + b3, 4)
ANSWER_B2a_occ_intercept = round(b0 + b2, 4)
ANSWER_B2a_unocc_slope = round(b1, 4)
ANSWER_B2a_unocc_intercept = round(b0, 4)
print(f"ANSWER_B2a_occ_slope = {ANSWER_B2a_occ_slope}")
print(f"ANSWER_B2a_occ_intercept = {ANSWER_B2a_occ_intercept}")
print(f"ANSWER_B2a_unocc_slope = {ANSWER_B2a_unocc_slope}")
print(f"ANSWER_B2a_unocc_intercept = {ANSWER_B2a_unocc_intercept}")
```

(B2-b, 6 pts) ASHRAE 90.1 minimum-efficiency provisions include a nighttime setback requirement: the HVAC system must be capable of reducing cooling output by at least 30% during unoccupied periods. At the 1% design condition (T = 108°F):

```python
ANSWER_B2b_occ_108 = round(ANSWER_B2a_occ_slope*108 + ANSWER_B2a_occ_intercept, 2)
ANSWER_B2b_unocc_108 = round(ANSWER_B2a_unocc_slope*108 + ANSWER_B2a_unocc_intercept, 2)
ANSWER_B2b_ratio = round(ANSWER_B2b_unocc_108 / ANSWER_B2b_occ_108, 4)
print(f"ANSWER_B2b_occ_108 = {ANSWER_B2b_occ_108}")
print(f"ANSWER_B2b_unocc_108 = {ANSWER_B2b_unocc_108}")
print(f"ANSWER_B2b_ratio = {ANSWER_B2b_ratio}")
```

In a markdown cell: does the unoccupied load represent less than 70% of the occupied load at this temperature (i.e., is ANSWER_B2b_ratio < 0.70)? State your conclusion about whether this satisfies the setback intent.

(B2-c, 6 pts) Compute the occupancy load fraction at three outdoor temperatures:

$$\text{Occupancy fraction} = \frac{\hat{y}_{\text{occupied}} - \hat{y}_{\text{unoccupied}}}{\hat{y}_{\text{occupied}}}$$

```python
def occ_fraction(T):
    occ = ANSWER_B2a_occ_slope*T + ANSWER_B2a_occ_intercept
    unocc = ANSWER_B2a_unocc_slope*T + ANSWER_B2a_unocc_intercept
    return (occ - unocc) / occ

ANSWER_B2c_frac_70 = round(occ_fraction(70), 4)
ANSWER_B2c_frac_90 = round(occ_fraction(90), 4)
ANSWER_B2c_frac_108 = round(occ_fraction(108), 4)
print(f"ANSWER_B2c_frac_70 = {ANSWER_B2c_frac_70}")
print(f"ANSWER_B2c_frac_90 = {ANSWER_B2c_frac_90}")
print(f"ANSWER_B2c_frac_108 = {ANSWER_B2c_frac_108}")
```

In a markdown cell: does the occupancy fraction increase, decrease, or remain approximately constant as outdoor temperature rises? Explain physically why: as outdoor temperature increases, does the incremental effect of occupancy become a larger or smaller fraction of total cooling demand?

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## Problem 3 — Written Reflection (12 pts, written)

Write a 100–150 word reflection (no code required) in a markdown cell, addressing the following prompt:

> The interaction model ($R^2$ = 0.795) outperforms the additive model ($R^2$ = 0.697) at the cost of greater complexity and a harder-to-explain coefficient structure. Describe one real-world scenario where you would recommend the simpler additive model despite its lower $R^2$, and one scenario where the interaction model is the better choice. Support each recommendation with a specific quantitative observation from the lab (e.g., a coefficient value, an $R^2$ comparison, a prediction difference, or a slope difference).

Grading criteria (12 pts total): - Additive model recommendation: scenario is specific and plausible, supported by at least one lab number (4 pts) - Interaction model recommendation: scenario is specific and plausible, supported by at least one lab number (4 pts) - Writing quality: concise, organized, quantitative evidence used appropriately, within 100–150 words (4 pts)

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## Submission Checklist

- ☐ Setup cell run with MAJOR declared
- ☐ P1a: two ANSWER_ season means printed + written response ~75 words using "collinearity"
- ☐ P1b: model summary shown; ANSWER_ r2/coef/pvalue printed; written comparison to Model 2
- ☐ P1c: model summary shown; ANSWER_ r2/adj_r2 printed; written conclusion about DayType's incremental information
- ☐ P1d: four Model 5 coefficients stated; ANSWER_ predictions printed; written comparison to Model 2
- ☐ P2 (only your declared track): all ANSWER_ values printed and written responses
- ☐ P3: reflection, 100–150 words, two scenarios, two quantitative observations